LEF MODEL · CHAPTER 18

Chapter 18: The Convergence of the Whole Pacific, the Spatula Ridges and the Solution to the Riddle of the Floor's Young Age

Based on Paper 09 — The Stripes: The leading side of the pear on the floor of the Pacific (Parts C.5–H: the GEBCO 2026 tracing, the spatula ridges, the ancient equator, the 3 challenges and the 10 studies)

01 · THE MYSTERY

The mystery: do all the lines of the Pacific lead to the tip of the pear — and how do stripes billions of years old survive in a floor less than 180 million years old?

In Chapter 17 we saw how the six parallel stripes of the North Pacific bound, between two knees, the mirror footprint of the Moon in the west. But to turn a visual insight into measurable science, Paper 09 (Parts C.5–F) raises three fundamental questions which, if not resolved, would leave the whole explanation a hypothesis:

1. **Do the stripes of the central and southern Pacific also converge on the same tip (C.5)?** South of the equator extend further giant systems — the Sala y Gómez–Easter Island–Tuamotu chain in the centre, and the Eltanin–Louisville–Tuvalu system in the far south. In a blind quantitative measurement on the global GEBCO 2026 depth grid, do all ten of these tracks converge on the final contact zone of the Moon (8.7°N, 167.7°E)?

2 **Why do the giant island chains of the Marshalls and the Line Islands have no mantle-plume age progression (D.1–D.3)?** In the mainstream tectonic model, island chains in the Pacific form when an oceanic plate moves over a fixed "hotspot" in the mantle — so the age must progress continuously from one end to the other (as in Hawaii–Emperor, 0 to 81 Ma). But in the Line Islands and the Marshall Islands the ages jump without order, erupting simultaneously along thousands of kilometres!

3 **The geochronological age paradox (F.1):** the Pacific floor between the stripes is dated (by magnetic anomalies and basalts) to 180 million years at most — billions of years younger than the moment of the Moon's detachment (3.873 Ga). How can stripes created when the pear was drawn back in be imprinted in a floor that formed billions of years later?

02 · AN IMAGE

An image: fresh snow on a ploughed field, plaster drawn under a spatula, and a lump of plasticine pushed into a spinning ball

Three simple, tangible images solve these three questions and join them into one physical picture:

- **Fresh snow falling on a ploughed field (solving the riddle of the ocean floor's young age, F.1):** imagine a field ploughed into deep furrows months ago, and last night a light snow fell on it. The fresh snow wraps the furrows and keeps showing the lines of ploughing through the white blanket. If you date the snow, you get the date of last night's snowstorm — not the date of the ploughing! In the same way, the ocean floor accumulates new layers from above while the layers beneath are renewed from below: radiometric dating measures only **the last top layer (< 180 Ma)**, while the stripes are **deep density folds** in the lower layer that the young layers have wrapped — which is why the stripes are faint in the outer topography but remarkably sharp on the gravity-gradient map (VGG), which sees into the depth!

- **A plasterer pulling a flat spatula off soft plaster ("the Spatula Ridges", D.1):** press a flat spatula against a wall smeared with soft plaster and pull it away hard: the plaster sticks to the spatula and to the wall, stretches between them, and finally tears into two **raised ridges along the edges of the spatula**. When the Moon peeled and pulled itself out hard beyond the corotation line (as the contact zone shrank from 6,000 to 3,000 km), the soft material of the contact zone was drawn upward into two chains of ridges — **the Marshall–Gilbert–Tuvalu chain in the west and the Line Islands chain in the east** (2,800–3,500 km apart!) — which came down to the ocean floor together with the tip when the pear was drawn back in!
- **A heavy lump of plasticine pushed from the equator into the depths of a spinning ball (the tilt of the stripes' equator, E.3):** stick a heavy lump of plasticine (7–12 lunar masses) on the equator of a spinning ball; the ball will spin stably. Now push the lump inward into the depths of the ball: the moment of inertia changes, the balance is disturbed, and the ball slowly rolls relative to its rotation axis until it finds a new equilibrium — which is why the ancient equator recorded in the stripes is tilted today by **≥ 10°** relative to today's equator!

03 · SCIENCE AND MODEL

Science and model: the GEBCO 2026 tracing across the whole Pacific and the age patterns of the spatula ridges (Parts C.5–E)

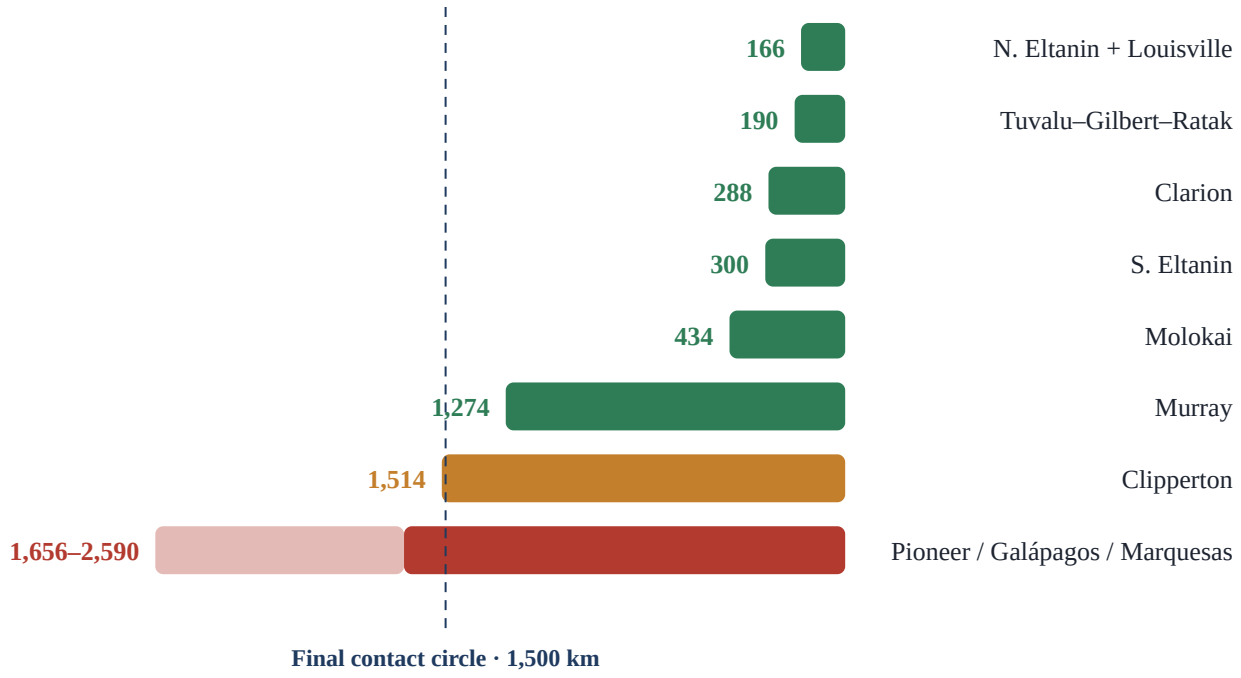
Parts C.5–E of Paper 09 present an unprecedented quantitative test of all the line systems in the Pacific:

1. **The convergence study across the whole Pacific on the GEBCO 2026 grid (C.5):** the author defined in advance, in writing, an algorithmic protocol (a directional filter in steps of 0.5° on the GEBCO 2026 shaded-relief map at a resolution of 30 points per degree) and two fixed targets: **the final contact** (8.7°N, 167.7°E, radius 1,500 km) and **the wide tip zone** (6°N, 179.5°W, radius 3,000 km). The results:

- 2 **All 10 traced tracks** (from the north, the centre and the south) reach into the tip zone (3,000 km).
- 3 **6 of the 10 tracks** (five of them within only 450 km!) pass deep inside the circle of the final contact (1,500 km): the upper southern track (northern Eltanin + Louisville) at **166 km**, the Tuvalu–Gilbert–Ratak chain entering the contact directly at **190 km**, Clarion at **288 km**, the lower southern track (southern Eltanin) at **300 km**, Molokai at **434 km**, and Murray at 1,274 km (and Clipperton misses by only 14 km — 1,514 km).
- 4 **0 of the 3 control lines** (Pioneer, Galápagos and Marquesas, which were not marked as tracks to the tip) enter the circle of the final contact (1,656–2,590 km)!

5 **The spacing of the spatula ridges and the absence of a mantle-plume age progression (D.1–D.3):** the distance between the Marshall–Gilbert–Tuvalu chain in the west and the Line Islands chain in the east is **2,800–3,500 km** (Majuro to Kiritimati 3,490 km, Tarawa to Palmyra 2,810 km, middle to middle 3,290 km) — identical to the width of the contact zone (3,000 km) and to the distance between the two knees (3,360 km). Moreover, the pair of chains is shifted about half a width east toward the base (Kiritimati sits on the Line Islands, and Kwajalein in the Marshalls passes 180 km from the centre of the tip), because the spatula ridges were drawn upward within **the peeling band** between 6,000 and 3,000 km! And in the age test: in the Line Islands two simultaneous eruptions were measured along 1,200 to >4,000 km (81–86 Ma and 68–73 Ma) with no age progression at all (and in the literature they are attributed to lithospheric stretching!), and in the Marshall Islands ages of 66–140 Ma were measured with no plume order.

6 **The tilt of the stripes' equator (≥ 10°, Part E):** measuring the slope of each stripe relative to today's latitude lines reveals that all the stripes climb north as they go east: Murray (+12.7°), Clarion (+9.0°), Clipperton (+11.0°), the southern stripe (+11.7°) and Mendocino (+2.4°). A joint fit requires an equator tilted by **at least 10°** (and up to ~ 20°; in the 11° fit it crosses today's equator at 159°W, only 1.6° from Kiritimati).



The distance of the last point of each (extended) track from the centre of the final contact, in the GEBCO 2026 tracing: six tracks penetrate the 1,500 km circle, none of the control lines does.

04 · THE MECHANISM IN WORDS

The mechanism in words: answering the three geological challenges (Part F)

Part F of Paper 09 faces head-on the three main challenges geologists raise against the interpretation of the stripes and the spatula ridges, and defines for each of them the decisive physical test:

1. **Challenge 1 — the young age of the ocean floor (< 180 Ma, §F.1):** the Pacific floor between the stripes is dated by magnetic anomalies and basalts to 180 million years at most. In the LEF model, the ocean floor is not "one block" created in one moment, but **a crust renewed in layers**: new layers accumulate from above while the layers beneath are taken and renewed from below. Radiometric dating measures only **the last top blanket** (like dating snow that fell last night on a ploughed field), while the stripes are **deep density folds** left in the layers beneath! That is why the stripes are blurred in the outer topography but remarkably sharp on the gravity-gradient map (VGG), which exposes density contrasts at depth (and which was in any

case described in Sandwell et al. 2014 as revealing "buried tectonic structure"). **The test (Study 9):** in seismic profiles and boreholes crossing the stripes, the fold will continue beneath the young layers, and the layers above it will be thinner over the crests of the fold and thicker in its troughs.

2. **Challenge 2 — the order of formation of the stripes and the ridges (§F.2):** the fact that the stripes pass beneath the volcanic ridges without cutting them (B.5) proves that the stripes formed by folding from below and from the sides, but does not by itself determine which came first — the ridges may already have stood on the floor while the fold formed beneath them (like a table standing on a tablecloth that crumples beneath it). **The test (Study 6):** precise coordinate measurement on a flat map without vertical exaggeration will check whether the ridges themselves underwent **shortening, compaction or bending** exactly at the points where the stripes cross beneath them.

3. **Challenge 3 — the orderly age progression of the Hawaii–Emperor chain (§F.3):** unlike the Line Islands (81–86 and 68–73 Ma simultaneously) and the Marshall Islands (66–140 Ma with no order), the Hawaii–Emperor chain shows a clean age progression from 0 Ma in Hawaii to ~ 81 Ma at the Emperor end, consistent with a classic mantle plume. The author makes clear that **the theory does not rest on Hawaii–Emperor** (offered only as a secondary hypothesis), but stands on the Line Islands and the Marshalls — and is a candidate for a controlled comparison against plume ridges outside the Pacific (Ninetyeast, Walvis and Chagos–Laccadive) on five measures: ages, morphological shape, flexural moat, crustal root and direction (Study 7).

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The measurement of the tilt angle θ of each stripe relative to today's latitude lines on a sphere, and the conservation of the Earth's moment of inertia I as the pear's mass (7–12 lunar masses) was drawn from the surface into the depths of the lower mantle, are formulated as follows (§E.1, §E.3):

$$\tan\theta = \frac{\Delta\phi}{\Delta\lambda\cos\bar{\phi}} \geq \tan(10^\circ)$$

$$I = \int r_\perp^2 \, dm \implies \Delta I_{\text{pear}} \rightarrow \text{Axial Roll } (\geq 10^\circ)$$

Since moving a mass of $7\text{–}12\ M_{\text{Moon}}$ from a distance of thousands of kilometres above the equator to the base of the mantle (the Pacific LLVP) changes the Earth's inertia tensor, the Earth slowly rolled until it found a new rotational equilibrium — leaving the equator of the stripe system tilted by ≥ 10° relative to the modern equator.

06 · THE SIGNATURE

The signature in reality: the ten studies of Part G and the traffic-light summary of Part H

In the summary of Paper 09 (Parts G–H), **6 verified findings ("green light")** and **4 observations awaiting the protocol of the 10 quantitative studies ("orange light")** are presented:

- **The six verified anchors ("green light"):** (1) the base of the pear at Kiritimati sits 1.6° from the stripes' equator, next to Clipperton (1.8°); (2) the width of the tip between the knees (3,360 km) matches the contact (3,000 km) and the dome (3,650 km); (3) the length of the stripe system (9,360 km) matches the length of the pear's body (8,890–9,000 km); (4) the mirror imprint at × 1.37 places Imbrium, Serenitatis and the contact centre within the footprint; (5) the stripes pass continuously beneath the ridges at four intersections; and (6) in the GEBCO 2026 tracing all 10 tracks of the north, the centre and the south reach the tip zone, and 6 of them penetrate the final contact (1,500 km), against 0 of the control lines.
- **The four directions for further research ("orange light"):** quantifying the crumpled arcs on the back side (8,100–10,300 km), measuring the convergence angles on a flat map without vertical exaggeration (Study 6), comparing the spatula ridges on five measures against plume ridges

worldwide (Study 7), and examining seismic profiles and boreholes beneath the young floor layer (Study 9).

The lab box: results of the GEBCO 2026 tracing (C.5), the spatula ridges (D) and the 10 studies (Paper 09, Parts C.5–H)

System / track in GEBCO 2026	Last reliable point and direction	Distance from the centre of the final contact (8.7°N, 167.7°E)	Distance from the centre of the wide tip (6°N, 179.5°W)	Pre-defined verdict
Upper south: Eltanin (northern branch) + Louisville	**174.6°W, 26.8°S (NNW 335°)**	**166 km (extension 4,380 km)**	**1,025 km**	**Inside the final contact (< 1,500 km)**
Upper south beyond Fiji: Tuvalu–Gilbert–Ratak	**167.9°E, 10.7°N (NW 319°)**	**190 km (0 km extension — enters directly!)**	**851 km**	**Inside the final contact (< 1,500 km)**
North: Clarion	**154.3°W, 13.8°N (W 270°)**	**288 km (extension 4,170 km)**	**760 km**	**Inside the final contact (< 1,500 km)**
Lower south: Eltanin (southern branch)	**154.2°W, 52.5°S (NW 325°)**	**300 km (extension 7,690 km)**	**928 km**	**Inside the final contact (< 1,500 km)**
North: Molokai	**169.7°W, 18.9°N (WSW 257°)**	**434 km (extension 2,660 km)**	**1,147 km**	**Inside the final contact (< 1,500 km)**
North: Murray	**168.6°W, 27.0°N (W 260°)**	**1,274 km (extension 2,980 km)**	**2,004 km**	**Inside the final contact (< 1,500 km)**
North: Clipperton / Mendocino	165.7°W, 0.1°S / 169.1°W, 36.8°N	1,514 km / 1,826 km	944 km / 2,770 km	Inside the tip zone (< 3,000 km)
Centre: Sala y Gómez–Easter–Tuamotu–Line (+ western branch)	169.7°W, 16.4°N / 171.6°W, 11.2°N	2,441 km / 2,094 km	1,601 km / 1,340 km	Inside the tip zone (< 3,000 km)

3 control lines (Pioneer, Galápagos, Marquesas)	Not marked as tracks to the tip	**1,656–2,590 km**	1,769–2,563 km	**0 of 3 enter the final contact**

Ridge chain (Part D)	The measured age pattern	Interpretation in the LEF model
The Line Islands (eastern spatula ridge)	**Two synchronous episodes (81–86 and 68–73 Ma) along 1,200–4,000+ km, with no age progression**	**Drawn upward in the eastern peeling band (2,800–3,500 km from the Marshalls); passes through Kiritimati**
Marshall–Gilbert–Tuvalu (western spatula ridge)	**66–140 Ma, with no simple age progression, fragmented eruptions**	**Drawn upward in the western peeling band; passes through the centre of the tip (Kwajalein 180 km from the centre)**
Hawaiian–Emperor	A clean progression from 0 Ma to ~ 81 Ma	Offered only as a secondary hypothesis (not required); to be tested in Study 7 against Ninetyeast, Walvis and Chagos–Laccadive

07 · NEXT

The door ahead

Chapter 19

Young Mars — the Pointed Bulge of Tharsis and the Ejection of Deimos and Phobos from Pavonis Mons →

← Chapter 17 · Prologue

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